acceleration at centre of
gravity
TABLE 1
Value
0.20 g~/
0.35 g
Ref. 4.3.3
Ref. 4.3.1
COMMENTS
0.08 g and 0.20 g/s~/: Elderly person will
keep balance when
holding
0.15 g and 0.20 g/s: Mean person will keep
balance when holding
0.15 g and 0.80 g/s: Sitting person will
start holding
0.25 g and 2.0 g/s: Maximum load for mean
person keeping balance
when holding
0.45 g and 10 g/s: Mean person falls out
of seat when not
wearing seat belts.
Risk of injury to passengers, safe emergency
operation after collision.
1.0 g: Degradation of passenger safety.
Loss of craft or/and
fatalities.

---CHUNK---

SOURCE: MSC.36(63).pdf
when holding
0.45 g and 10 g/s: Mean person falls out
of seat when not
wearing seat belts.
Risk of injury to passengers, safe emergency
operation after collision.
1.0 g: Degradation of passenger safety.
Loss of craft or/and
fatalities.
~I The recording instruments used should be such that acceleration accuracy is
value and frequency response should be minimum 20 Hz. Antialiasing filters
attenuation 100 + 5% should be used.
better than 5% of the real
with maximum passband
~I
~/
g = gravity acceleration (9.81 m/s2)
g-rate or jerk may be evaluated from acceleration/time curves.
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RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
ADOPTION OF THE INTERNATIONAL CODE OF SAFETY
FOR HIGH SPEED CRAFT TABLE 2
SAFETY LEVEL 1 1 1 2 3 4
EFFECT ON Normal Nuisance Operating . Emergency procedures; Large reduction Deaths,
CRAFT AND limitations significant reduction in safety margins; usually
OCCUPANTS in safety margins; crew over-burden with loss
difficult for crew to because of work- of craft
cope with adverse load or environ-
conditions; passenger mental conditions;

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SOURCE: MSC.36(63).pdf
OCCUPANTS in safety margins; crew over-burden with loss
difficult for crew to because of work- of craft
cope with adverse load or environ-
conditions; passenger mental conditions;
injuries. serious injury to a
small number of
occupants.
F.A.R . .ll EXTREMELY
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(reference) IMPROBABLE
only)
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IMPROBABLE
REASONABLY EXTREMELY
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PROBABILITY PROBABLE REMOTE
10- 0 10- 1 10- 2 10- 3 10- 4 10- 5 10- 6 10- 7 10- 8 10- 9
CATEGORY
OF EFFECT .- MINOR ~MAJOR) f-- HAZARDOUS) CATASTRO-- ,
PHIC
.11 The United States Federal Aviation Regulations
~/ European Joint Airworthiness Regulations
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RESOLUTION MSC.36(63)
(adopted on 20 May 1994)
ADOPTION OF THE INTERNATIONAL CODE OF SAFETY
FOR HIGH SPEED CRAFT ANNEX 4
MSC 63/23/Add.2
ANNEX 18
Page 185
PROCEDURES FOR FAILURE MODE AND EFFECTS ANALYSIS
1 Introduction
1.1 In the case of traditional craft, it has been possible to specify certain aspects of design or construction in some level of detail, in a way which was consistent with some level of risk which had over the years been intuitively accepted without having to be defined.
1.2 With the development of large high speed craft, this required experience has not been widely available. However, with the now broad acceptance of the probabilistic approach to safety assessments within industry as a whole, it is proposed that an analysis of failure performance may be used to assist in the assessment of the safety of operation of high speed craft.
1.3 A practical, realistic and documented assessment of the failure characteristics of the craft and its component systems should be undertaken with the aim of defining and studying the important failure conditions that may exist.

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SOURCE: MSC.36(63).pdf
1.3 A practical, realistic and documented assessment of the failure characteristics of the craft and its component systems should be undertaken with the aim of defining and studying the important failure conditions that may exist.
1.4 This annex describes a failure mode and effects analysis (FMEA) and gives
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guidance as to how it may be applied by:
.1 explaining basic principles;
.2 providing the procedural steps necessary to perform an analysis;
.3 identifying appropriate terms, assumptions, measures and failure modes; and
.4 providing examples of the necessary worksheets.
1.5 FMEA for high speed craft is based on a single failure concept under which each system at various levels of a system's functional hierarchy is assumed to fail by one probable cause at a time. The effects of the postulated failure are analysed and classified according to their severity. Such effects may include secondary failures (or multiple failures) at other level(s). Any failure mode which may cause a catastrophic effect to the craft should be guarded against by system or equipment redundancy unless the probability of such failure is extremely improbable (refer to section 13).

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SOURCE: MSC.36(63).pdf
level(s). Any failure mode which may cause a catastrophic effect to the craft should be guarded against by system or equipment redundancy unless the probability of such failure is extremely improbable (refer to section 13).
For failure modes causing hazardous effects corrective measures may be accepted in lieu. A test programme should be drawn to confirm the conclusions